# Computational trade-offs in penalized least-squares trend estimation: normalized smoothness, numerical inversion, and solver reuse

Heriberto Espino-Montelongo[1*] and Daniela Cortés-Toto[1]

[1*]Universidad de las Américas Puebla, San Andrés Cholula,Puebla,Mexico.

**Abstract**

Penalized least-squares (PLS) trend smoothing has well-established statistical foundations, but the relative cost of selecting a smoothing parameter depends on the linear-algebra backend, reusability, and parameterization. We present a reproducible comparison organized around the normalized trace smoothness coordinate $\boldsymbol{S} \in [\mathbf{0}, \mathbf{1}]$, with arbitrary finite-difference order and a fixed or data-derived reference difference mean. The computational study distinguishes complete parameter searches from individual trend fits and $\boldsymbol{S}$-inversion calls. It combines a previously archived 80,640-trial comparison, a paired solver experiment with seed-level bootstrap intervals and multiplicity-adjusted tests, and a float64 comparison of Brent, safeguarded Newton, safeguarded secant, and bisection. End-to-end rankings vary with workload and preparation reuse; inversion performance additionally depends on distance to the smoothness boundaries. Frozen monthly atmospheric $CO_{\mathbf{2}}$ and federal funds observations illustrate CV, GCV, AICc, BIC, conditional Gaussian profile likelihood, and prescribed smoothness for four difference orders. All experiments and figures can be regenerated independently from archived measurements. The results support workload-specific numerical choices rather than a universally best solver or forecasting rule.

**Keywords:** penalized least squares, Whittaker–Henderson smoother, numerical algorithms, smoothness inversion, reproducible benchmarking, time series

# 1 Introduction

Trend smoothing is useful when a noisy observed sequence contains persistent behavior that must be isolated from short-term variation. The finite-difference penalized least-squares (PLS) model goes back to Whittaker and subsequent developments, including Guerrero's formulation with reference differences Whittaker (1923); Guerrero (2007). Selecting the penalty by CV, GCV, AICc, or BIC is well studied Cortés-Toto et al. (2017), as are structured algorithms for Whittaker–Henderson problems Weinert (2007). Our aim is *not* to claim the penalty or its spectral representation as new. Instead, we quantify the engineering trade-offs that occur when penalties must be repeatedly evaluated, smoothness must be specified on a common scale, and numerical results must be checked.

The experimental presentation follows the distinction between *runtime and quality* used in comparisons of inference algorithms, including Berns et al. Berns et al. (2022). Gauss et al. Gauss et al. (2021) provide a complementary example of evaluating smoothing choices against competing performance requirements, and Cardoso et al. Cardoso et al. (2025) exemplify an explicit algorithmic workflow and comparisons against alternatives. Those studies address different models; their numerical conclusions are *not* transferred to PLS.

Our contributions are: (i) a common trace-normalized smoothness coordinate and an explicit inversion-comparison protocol; (ii) matched-workload measurements of dense, spectral and banded backends, including paired uncertainty for full-curve evaluation; and (iii) a reproducible software and graphical study that preserves numerical outputs independently of figure rendering. Forecast extensions are illustrative, not a separate forecasting algorithm.

# 2 Penalized trends and computational coordinates

For equally spaced $y \in \mathbb{R}^n$, let $D_d \in \mathbb{R}^{(n-d)\times n}$ denote the $d$th forward-difference operator, with $1 \le d < n$. For a given $\lambda \ge 0$ and reference mean $\mu \in \mathbb{R}$,

$$\widehat{\tau}_{\lambda,\mu} = \arg\min_{\tau\in\mathbb{R}^n} \left\{ \|y-\tau\|_2^2 + \lambda \|D_d\tau - \mu\mathbf{1}\|_2^2 \right\}. \tag{1}$$

Writing $Q_d = D_d^\top D_d$, the unique finite-penalty solution is

$$\widehat{\tau}_{\lambda,\mu} = (I + \lambda Q_d)^{-1}(y + \lambda\mu D_d^\top \mathbf{1}). \tag{2}$$

Since $D_d$ has rank $n-d$, $Q_d$ has $d$ zero eigenvalues. The null space contains sampled polynomials of degree less than $d$. For positive eigenvalues $\delta_1, \ldots, \delta_{n-d}$ of $Q_d$, the smoother matrix $H_\lambda = (I + \lambda Q_d)^{-1}$ has $\operatorname{tr}(H_\lambda) = d + \sum_j (1+\lambda\delta_j)^{-1}$. Thus the normalized index

$$S(\lambda) = \frac{n - \operatorname{tr}(H_\lambda)}{n-d} = \frac{1}{n-d}\sum_{j=1}^{n-d} \frac{\lambda\delta_j}{1+\lambda\delta_j} \tag{3}$$

maps $[0, \infty]$ onto $[0, 1]$. For finite positive $\lambda$,

$$S'(\lambda) = \frac{1}{n-d} \sum_j \frac{\delta_j}{(1+\lambda\delta_j)^2} > 0.$$

Hence each $s \in (0, 1)$ has a unique $\lambda = S^{-1}(s)$. The exact endpoints are $S^{-1}(0) = 0$ and $S^{-1}(1) = \infty$. Evaluating the complementary expression for $1 - S$ avoids cancellation when $s$ approaches one.

For fixed $\mu$, the public implementation supports leave-one-out CV, GCV, AICc, BIC, conditional Gaussian profile likelihood (MLE), and controlled $S$. The first five optimize distinct objectives, while controlled $S$ is a *prescribed target*, not a criterion. When $\mu = \overline{D_d y}$ is used, the reported CV and MLE calculations condition on the data-derived reference; they do not refit $\mu$ inside each deletion or jointly maximize $(\lambda, \mu)$. Different selectors evaluated on the *same* finite grid are comparable as selection policies, but boundary optima do not establish global minimizers.

# 3 Numerical experiments

We separate *full curve reconstruction*, *scalar criterion evaluation*, *repeated parameter selection*, and *one inverse calculation*. Comparing timings across these tasks would be misleading: a reused SVD incurs a one-time factorization but enables many fast spectral evaluations; a banded Cholesky solve avoids that dense preprocessing for an isolated fit Weinert (2007).

## 3.1 End-to-end computation and statistical uncertainty

The archived author-run Windows factorial experiment comprises 80,640 recorded parameter-search trials, seven backends, five criteria, $d = 0, \ldots, 5$, two reference-mean settings, and distinct reuse scenarios. A second paired study compares full SVD, symmetric eigendecomposition, and *actual banded Cholesky* for end-to-end preparation plus full-trend fits. The latter uses independent simulated-data seeds and randomized technical timing blocks. One median per independent seed is the inferential unit. Geometric mean paired time ratios are obtained from seed-level log ratios, with 95% *pointwise* bootstrap intervals and two-sided sign-flip tests adjusted by Holm over the declared family. Technical repetitions are not independent observations; performance remains host- and workload-specific.

## 3.2 Numerical inversion of normalized smoothness

Using a prepared spectral smoother, we compare Brent's method Brent (1971), safeguarded Newton, safeguarded secant, and bisection on 11 target smoothness values from $10^{-14}$ through $1 - 10^{-14}$, lengths $n = 30, 60, 120$ and orders $d = 0, 1, 2, 3$. Per-target randomized complete blocks record latency, iterations, function evaluations, convergence failures and relative *tail* error. The $d = 0$ analytic inverse is excluded from numerical-root speed comparisons. These timings exclude initial spectral factorization; speed cannot replace an accuracy or convergence check.

**Table 1** Measured paired runtime ratios for SVD over banded Cholesky. Values above one favor banded Cholesky, and below one favor SVD. The displayed intervals are pointwise; Holm $p$ values refer to the recorded multiple-comparison family. $M$ denotes the number of fitted penalties.

| $n$ | $d$ | $M$ | SVD / BChol | 95% pointwise CI | Holm $p$ |
|---|---|---|---|---|---|
| 32 | 2 | 1 | 1.55 | [1.54, 1.57] | 0.00315 |
| 32 | 2 | 16 | 0.68 | [0.68, 0.68] | 0.00315 |
| 32 | 2 | 64 | 0.42 | [0.42, 0.42] | 0.00315 |
| 96 | 2 | 1 | 4.22 | [4.20, 4.24] | 0.00315 |
| 96 | 2 | 16 | 2.09 | [2.08, 2.10] | 0.00315 |
| 96 | 2 | 64 | 0.97 | [0.96, 0.97] | 0.00315 |

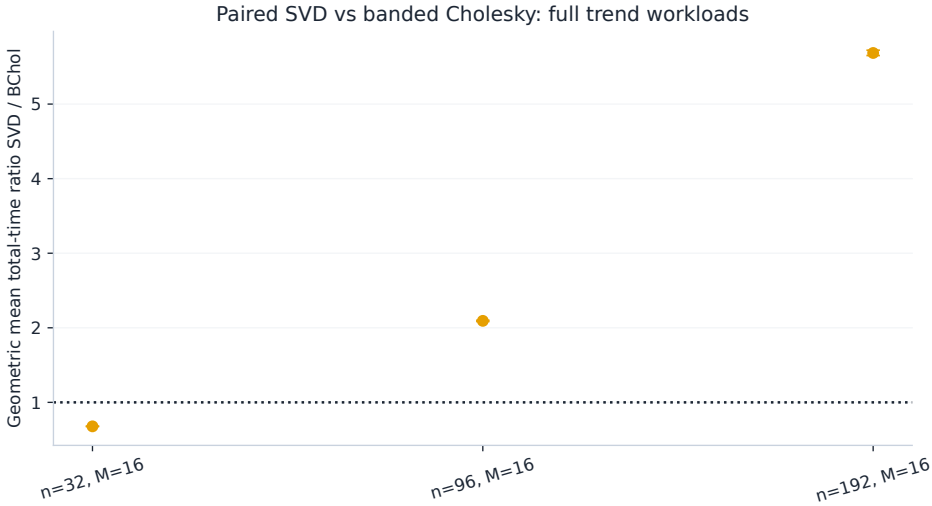



**Fig. 1** Paired solver speed ratios with seed-bootstrap intervals for the selected full-trend workloads. The horizontal reference denotes equal end-to-end time.

**Table 2** Measured inversion costs for two target smoothness values; the full grid and failure records are archived. Time is in microseconds and the last column checks the smaller smoothness tail.

| $S$ | Inversion method | Median $\mu$s | Fail / trials | Iterations | Max rel. tail error |
|---|---|---|---|---|---|
| .5 | brentq | 91.50 | 0/35 | 10.0 | 1.8e-15 |
| .5 | newton | 88.30 | 0/35 | 5.0 | 2.2e-16 |
| .5 | secant | 157.70 | 0/35 | 24.0 | 1.1e-15 |
| .5 | bisect | 292.00 | 0/35 | 45.0 | 1.4e-13 |
| .99 | brentq | 93.70 | 0/35 | 11.0 | 3.3e-14 |
| .99 | newton | 118.60 | 0/35 | 8.0 | 0 |
| .99 | secant | 162.80 | 0/35 | 26.0 | 1.7e-13 |
| .99 | bisect | 282.00 | 0/35 | 45.0 | 7.7e-14 |

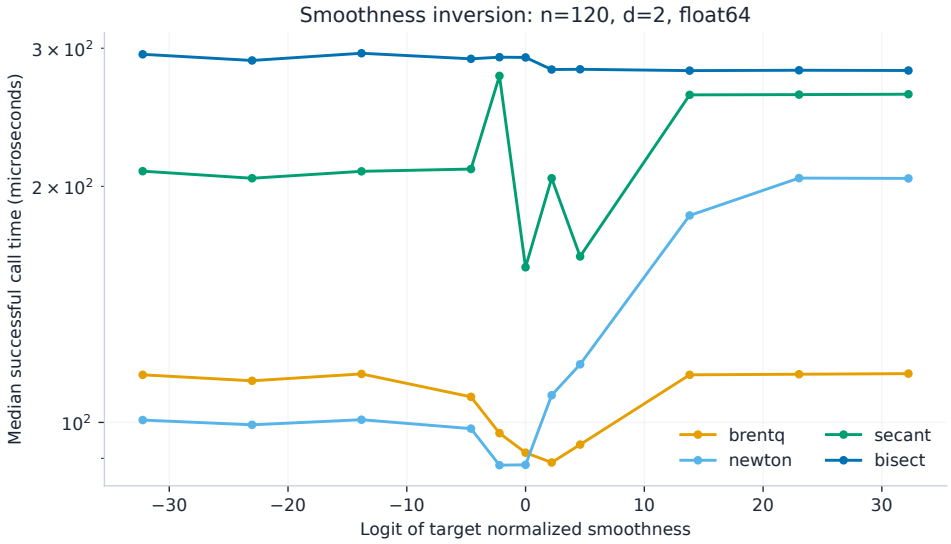



**Fig. 2** Measured float64 inversion time across logit($S$) for the four root finders. The source archive also reports every convergence failure and the actual attained smoothness accuracy.

## 4 Case study: contemporary monthly data

Two frozen public series provide contrasting use cases: 441 monthly Mauna Loa $CO_2$ observations (January 1990–September 2026) and 201 US monthly effective federal funds rates (January 2010–September 2026) NOAA Global Monitoring Laboratory and Scripps Institution of Oceanography (2026); Board of Governors of the Federal Reserve System (2026). For $d = 1, 2, 3, 4$ and $\mu = 0$ or $\mu = \overline{D_d y_{\text{train}}}$, we report selected $(\lambda, S)$ and a single 12-month historical holdout for CV, GCV, AICc, BIC, conditional MLE, and controlled $S = 0.86$. The 29 candidate $S$ values used by optimized selectors are the same within each configuration. Neither the selected $S$ nor a low in-sample score implies a better held-out extrapolation.

The continuation outside the observed window uses a separately specified locally fitted polynomial, anchored at the last smoothed value; PLS is a two-sided smoother and does not intrinsically forecast. $CO_2$ seasonality and policy-driven rate changes are not modeled, so the single-origin errors are diagnostics, not a statistically validated ranking of future forecasts.

## 5 Discussion and conclusion

The key engineering conclusion is conditional, not absolute: spectral reuse can be advantageous for repeated criterion calculations, while specialized banded solvers can be preferable for other reconstruction workloads. The measured solver ratios and inversion costs depend on data length, penalty order, number of evaluations, reuse, numerical tolerance, and host architecture. A normalized smoothness coordinate permits interpretable cross-order comparisons but introduces inversion costs.

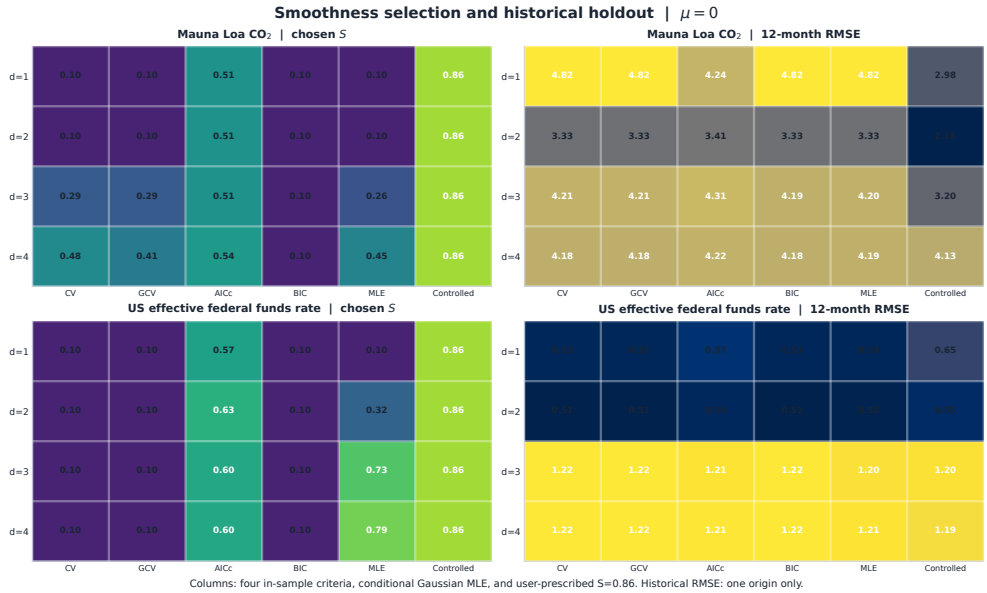



**Fig. 3** Compact archived-data dashboard: conditional selection and one-origin holdout errors by method and difference order on two monthly series. Rows and numerical values are from the saved result JSON, without retuning methods. Full $4 \times 6$ panels for each reference mean, including future trend-only extensions, are supplied online.

Observed selector differences do not, by themselves, demonstrate an improved forecasting methodology. The complete 80,640-trial factorial outputs, raw seed-level paired timings, full inversion targets, and high-resolution case dashboards are maintained as reproducible supplementary material rather than repeated in the main article.

## Declarations

**Funding and competing interests.** To be completed and verified by the coauthors before submission.

**Data availability.** Frozen NOAA and FRED observations, selection-result JSONs, timing CSVs, uncertainty analyses, and rendering manifests are archived in the companion repository. **Code availability.** https://github.com/heri-espino/trendestimation. A permanently archived tagged release should be cited on submission.

**Author contributions.** To be agreed by the coauthors.

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